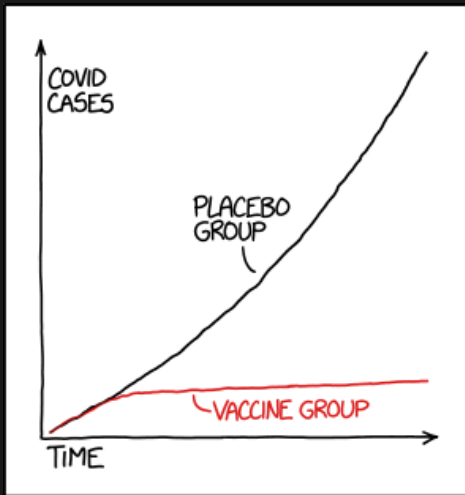


Cross Validation

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Lecture 02.2 (v4.0.0)



STATISTICS TIP: ALWAYS TRY TO GET DATA THAT'S GOOD ENOUGH THAT YOU DON'T NEED TO DO STATISTICS ON IT

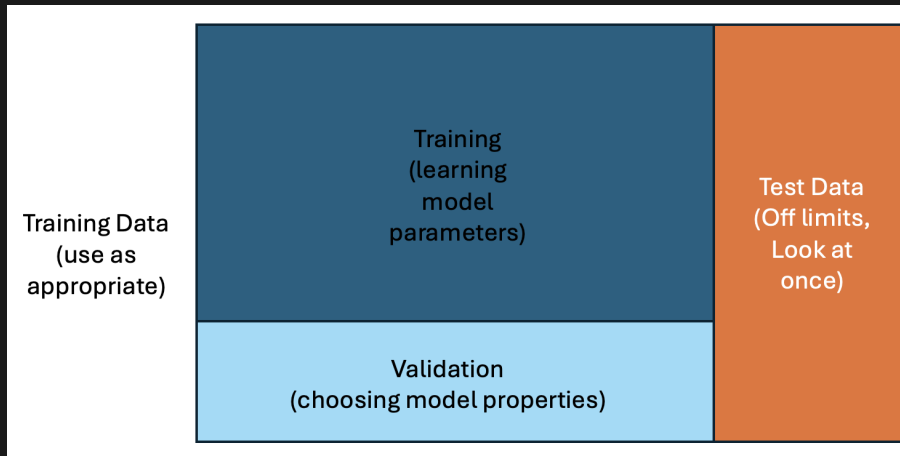
Signposting

- ▶ This lecture covers Cross Validation:
 - ▶ why you should always do it
 - ▶ What it looks like

Model Selection

- ▶ Imagine that we have run two different inference procedures (models) on our data.
- ▶ We want to decide which of these gives the **best** description of the data.
 - ▶ (we might pretend we want to know which one is **right**...)
- ▶ Model selection formalises how to make this assessment.
 - ▶ Focus on cross-validation: performance in held out data

Data distribution



K-fold CV

- ▶ **Split** the training data into k “folds” $f(i)$, that is, **random non-overlapping samples** of the data of size n/k . Then:
- ▶ **For each fold i :**
 - ▶ Call $X^{-(f(i))}$ the “training” dataset and $X^{(f(i))}$ the “validation” dataset
 - ▶ Learn parameters $\hat{\theta}_i$ with data $X^{-(f(i))}$
 - ▶ Evaluate $l_i = \text{Loss}(X^{(f(i))} | \hat{\theta}_i)$
- ▶ And report $\frac{1}{n} \sum_{i=1}^k l_i$

How to use the data?

- ▶ We can make many different CV splits of our **training** data
- ▶ To evaluate variance etc
- ▶ We can choose model structure, even do model choice with it
- ▶ When we look at the test data, that is it: report performance and stop
- ▶ Test data can be used to score, not to influence, the algorithm

How many folds?

- ▶ k-fold CV loses a fraction of the data
 - ▶ A **bias-variance tradeoff**: we introduce a bias (towards simpler models) but also significantly reduce the variance of the MSE estimation.
 - ▶ This means that (under the assumption that the **true model is not in the model space**) k-fold CV will choose a **simpler model** with less predictive power than was possible.
- ▶ However, smaller k can make the inference more consistent across different data.
- ▶ For **larger data**, $k = 10$ is often chosen:
 - ▶ **cost**. k defines the minimum number of times you need to run the models. If you can afford to run a model once, you can probably afford 10 times.
 - ▶ **practicality**. If you had only 10% more data you might expect to get the same performance as using validation data for training. We frequently lose this amount of data to quality control, etc.

Handling correlation

- ▶ **Correlation** structures can be handled in k-fold CV by **careful sampling**:
 - ▶ a-priori there is a correlation in time or space expected. we can therefore **remove windows**.
 - ▶ the data have some associated covariate, which can be removed en-masse.
 - ▶ empirical correlation structures can be used to select a point i and all points correlated with it above some **correlation threshold**.
- ▶ Some of these can be used in other contexts. Examples include:
 - ▶ **block bootstrap**
 - ▶ Using a different definition of a “datapoint” in a leave-one-out context, for example: datapoints are countries instead of countries at timepoints

General considerations

- ▶ To make Cross-Validation work, we need to be able to define our inference goal cleanly. Some scenarios:
 - ▶ **Same source, single datapoint**: Within a single datastream, how well can we predict the **next** point?
 - ▶ **Same source, segment of data**: Within a single datastream, how well could we predict everything that happens within an hour?
 - ▶ **New but understood source**: We have multiple datastreams, each of which might be different but all are generated by a similar process. How well can we predict a new such datasource?
 - ▶ **Unexpected source**: We have many classes of datastream. How well can we predict what would happen on a new class of datastream?

Reflection

- ▶ You should understand how to:
 - ▶ Define and use a null hypothesis significance test,
 - ▶ Contrast classical and resampling tests, and judge appropriate uses,
 - ▶ Use statistical testing appropriately in projects.

Further reading

- ▶ Cross Validation
 - ▶ Chapters 2.3 and 7.10 of [The Elements of Statistical Learning: Data Mining, Inference, and Prediction](#) (Friedman, Hastie and Tibshirani).
 - ▶ [Cosma Shalizi's Modern Regression Lectures](#) (Lectures 20, 26)